

Total No. of Questions : 8]

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Roll No

CE-8003(1)-CBGS

B.E. VIII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Pre-Stress Concrete Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Draw neat diagram.

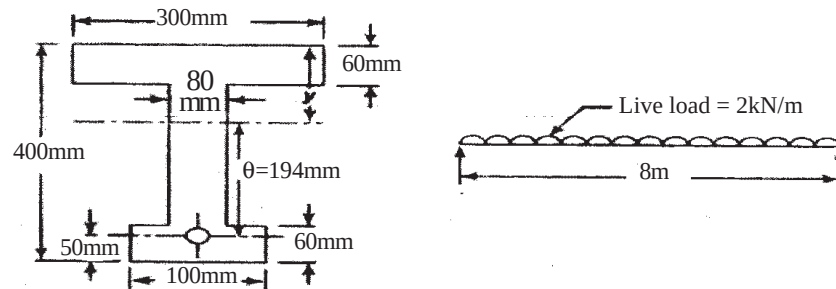
1. a) What is Prestressed concrete? Explain the concept of prestressing.
b) Explain Load Balancing Concept with neat sketch.
2. a) Explain why high strength concrete and high strength steel are needed for PSC construction.
b) Explain in detail, What is mean by Flexural strength.
3. An unsymmetrical I-section beam is used to support an imposed load of 2kN/m over a span of 8m. The sectional details are top flange, 300mm wide and 60mm thick; bottom flange, 100mm wide and 60mm thick; thickness of web=80mm; overall depth of the beam = 400mm. At the centre of the span, the effective pre-stressing force of 100kN is located at 50mm from the soffit of the beam. Estimate stresses at the centre of span section of the beam for following load conditions:
a) Prestress + Self weight

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b) Prestress + Self weight + Live load



4. A cylindrical concrete water tank of internal diameter 30m is required to store water over a depth of 7.5m. The permissible compressive stress in concrete at transfer is 13N/mm^2 and the minimum compressive stress under working pressure is 1N/mm^2 . The loss ratio is 0.75. Wires of 5mm diameter with an initial stress of 1000N/mm^2 are available for circumferential winding and freyssinet cables made up of 12 wires of 8mm diameter stressed to 1200N/mm^2 are to be used for vertical pre-stressing. Design the tank walls assuming the base as fixed. The cube strength of concrete is 40N/mm^2 .
5. Write down the general design aspects of pretensioned pre-stressed bridge deck.
6. A post-tensioned pre-stressed concrete slab deck has to be designed for a national highway crossing. Design the deck slab using the following data: Clear span = 10m, Width of bearing = 400mm, Clear width of road way = 15m, Thickness of wearing coat = 80mm, live load: IRC Class AA tracked vehicle, type of structure: I type, materials: M-40 grade concrete and cable containing 12 high tensile wires of 7mm diameter initially stressed to 1200N/mm^2 and Fe-415 HYSD bars compressive strength of concrete at transfer = 35N/mm^2 percentage loss of pre-stress = 20 percent. Design the deck slab and determine the spacing of concrete.

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7. Briefly explain losses in pre and post-tensioning systems.
8. Write a note on
 - i) Freyssinet System
 - ii) Magnel blaton system
 - iii) Hoyer system
 - iv) Gifford udall system
